

- 13.** A discount series of 10%, 20% and 40% is equal to a single discount of
(a) 50% (b) 60% (c) 56.8% (d) 70.28%
- 14.** Successive discounts of $12\frac{1}{2}\%$ and $7\frac{1}{2}\%$ are given on the marked price of a cupboard. If the customer pays ₹ 2590, then what is the marked price?
(a) ₹ 3108 (b) ₹ 3148 (c) ₹ 3200 (d) ₹ 3600
- 15.** The difference between a discount of 40% on ₹ 1000 and two successive discounts of 35% and 5% on the same amount is
(a) ₹ 15.50 (b) ₹ 16.50 (c) ₹ 17.50 (d) ₹ 18.00
- 16.** A dealer buys an article listed at ₹ 100 and gets two successive discounts of 10% and 20%. He spends 10% of the cost price on transport etc. At what price should he sell the article to earn a profit of 15%?
(a) ₹ 90 (b) ₹ 91 (c) ₹ 91.08 (d) ₹ 91.10
- 17.** An item costing ₹ 200 is being sold at 10% loss. If the price is further reduced by 5%, then the selling price will be
(a) ₹ 170 (b) ₹ 171 (c) ₹ 175 (d) ₹ 179
- 18.** A man bought a number of oranges at 3 for a rupee and an equal number at 2 for a rupee. At what price per dozen should he sell them to make a profit of 20%?
(a) ₹ 4 (b) ₹ 5 (c) ₹ 6 (d) ₹ 7
- 19.** A milk vendor bought 28 L of milk at the cost of ₹ 8.50 per L. After adding some water, he sold the mixture at the same price. If he gains 12.5%, then how much water did he add?
(a) 5.5 L (b) 4.5 L
(c) 3.5 L (d) 2.5 L
- 20.** A man sold two watches, each for ₹ 495. If he gained 10% on one watch and suffered a loss of 10% on the other, then what is the loss or gain percentage in the transaction?
(a) 1% gain (b) 1% loss
(c) 100/99% loss (d) No gain no loss
- 21.** Jyoti bought a computer system for ₹ 40000. She sold it to Brajesh at a loss of 4%. If Brajesh sells it for ₹ 40320 to Yash, then the profit per cent earned by Brajesh is
(a) 3% (b) 5%
(c) 7% (d) 10%
- 22.** List price of a video cassette is ₹ 100. A dealer sells three video cassettes for ₹ 274.50 after allowing discount at certain rate. The rate of discount allowed is
(a) 7% (b) 7.5%
(c) 8% (d) 8.5%
- 23.** A fruit-seller buys lemons at 2 for a rupee and sells them at 5 for three rupees. What is his gain per cent?
(a) 10% (b) 15% (c) 20% (d) 25%
- 24.** A trader marks 10% higher than the cost price. He gives a discount of 10% on the marked price. In this kind of sales how much per cent does the trader gain or loss?
(a) 5% gain (b) 2% gain (c) 1% loss (d) 3% loss
- 25.** One saree was purchased for ₹ 564 after getting a discount of 6% and another saree was purchased for ₹ 396 after getting a discount of 1%. Taking both the items as a single transaction, what is the percentage of discount?
(a) 3.5% (b) 4% (c) 7% (d) 7.5%
- 26.** A dishonest dealer professes to sell his good at cost price but uses a false weight and thus gains 25%. For a kilogram he uses a weight of
(a) 700 g (b) 750 g (c) 800 g (d) 850 g
- 27.** A person bought two old scooters for ₹ 9000. By selling one at a profit of 25% and the other at a loss of 20%, to neither gain nor loses. The cost of each scooter is
(a) ₹ 3500, ₹ 500 (b) ₹ 4500, ₹ 4500
(c) ₹ 4000, ₹ 5000 (d) ₹ 5300, ₹ 3700
- 28.** The manufacturer of a certain item can sell all he can produce at the selling price of ₹ 60 each. It costs him ₹ 40 in materials and labour to produce each item and he has overhead expenses of ₹ 3000 per week in order to operate the plant. The number of items he should produce and sell in order to make a profit of atleast ₹ 1000 per week is
(a) 400 (b) 300 (c) 250 (d) 200
- 29.** A person sold a table at a profit of $6\frac{1}{2}\%$. If he had sold it for ₹ 1250 more, he would have gained 19%
I. The CP of the table is ₹ 10000.
II. CP will ₹ 19000, if he had sold it for ₹ 2400 more and gained same profit
Which one is correct?
(a) Only II (b) Only I
(c) Neither I nor II (d) Both I and II
- 30.** A bookseller sells a book at a gain of 10%. If he had bought it at 4% less and sold it for ₹ 6 more, he would have gained $18\frac{3}{4}\%$
I. The SP of the book is ₹ 165.
II. The CP of the book is ₹ 150.
Which one is correct?
(a) Neither I nor II (b) Both I and II
(c) Only I (d) Only II

> PREVIOUS YEARS' QUESTIONS

- 31.** A person bought 8 quintal of rice for certain rupees. After a week, he sold 3 quintal of rice at 10% profit, 3 quintal of rice with neither profit nor loss and 2 quintal at 5% loss. In this transaction, what is the profit? **☑ 2012 I**
(a) 10% (b) 20% (c) 25% (d) None of these
- 32.** The cost of two articles are in the ratio 3 : 5. If there is 30% loss on the first article and 20% gain on the second article, then what is overall percentage of loss or gain? **☑ 2012 I**
(a) 2.25% gain (b) 5.25% loss
(c) 2% loss (d) None of these
- 33.** A cloth store is offering Buy 3, get 1 free. What is the net percentage discount being offered by the store? **☑ 2012 II**
(a) 20% (b) 25% (c) 30% (d) $33\frac{1}{3}\%$
- 34.** A person sold an article for ₹ 136 and got 15% loss. Had he sold it for ₹ x , he would have got a profit of 15%. Which one of the following is correct? **☑ 2012 II**
(a) $190 < x < 200$ (b) $180 < x < 190$
(c) $170 < x < 180$ (d) $160 < x < 170$
- 35.** A man buys a television set which lists for ₹ 5000 at 10% discount. He gets an additional 2% discount (after the first discount) for paying cash. What does he actually pay for the set? **☑ 2012 II**
(a) ₹ 4410 (b) ₹ 4400 (c) ₹ 4000 (d) ₹ 4500
- 36.** A merchant earns a profit of 20% by selling a basket containing 80 apples which costs ₹ 240 but he gives one-fourth of it to his friend at cost price and sells the remaining apples. In order to earn the same profit, at what price must he sell each apple? **☑ 2012 II**
(a) ₹ 3.00 (b) ₹ 3.60 (c) ₹ 3.80 (d) ₹ 4.80
- 37.** A person sold an article for ₹ 3600 and got a profit of 20%. Had he sold the article for ₹ 3150, how much profit would he have got? **☑ 2013 II**
(a) 4% (b) 5% (c) 6% (d) 10%
- 38.** Two lots of onions with equal quantity, one costing ₹ 10 per kg and the other costing ₹ 15 per kg, are mixed together and whole lot is sold at ₹ 15 per kg. What is the profit or loss?

☑ 2013 II

- (a) 10% loss (b) 10% profit
(c) 20% profit (d) 20% loss
- 39.** On a 20% discount sale, an article costs ₹ 596. What was the original price of the article? **☑ 2014 I**
(a) ₹ 720 (b) ₹ 735 (c) ₹ 745 (d) ₹ 775
- 40.** A man buys 200 oranges for ₹ 1000. How many oranges for ₹ 100 can be sold, so that his profit percentage is 25%? **☑ 2014 II**
(a) 10 (b) 14 (c) 16 (d) 20
- 41.** When an article is sold at 20% discount, the selling price is ₹ 24. What will be the selling price when the discount is 30%. **☑ 2014 II**
(a) ₹ 25 (b) ₹ 23 (c) ₹ 21 (d) ₹ 20
- 42.** A shopkeeper sells his articles at their cost price but uses a faulty balance which reads 1000 g for 800 g. What is his actual profit percentage? **☑ 2014 II**
(a) 25% (b) 20%
(c) 40% (d) 30%
- 43.** A person selling an article for ₹ 96 finds that his loss per cent is one-fourth of the amount of rupees that he had paid for the article. What can be the cost price? **☑ 2014 II**
(a) Only ₹ 160 (b) Only ₹ 240
(c) Either ₹ 160 or ₹ 240 (d) Neither ₹ 160 nor ₹ 240
- 44.** A milkman claims to sell milk at its cost price but he is making a profit of 20% since, he has mixed some amount of water in the milk. What is the percentage of milk in the mixture? **☑ 2015 I**
(a) 80% (b) $\frac{250}{3}\%$ (c) 75% (d) $\frac{200}{3}\%$
- 45.** The value of a single discount on some amount which is equivalent to a series of discounts of 10%, 20% and 40% on the same amount, is equal to **☑ 2015 II**
(a) 43.2% (b) 50% (c) 56.8% (d) 70%
- 46.** A cloth merchant buys cloth from a weaver and cheats him by using a scale which is 10 cm longer than a normal metre scale. He claims to sell cloth at the cost price to his customers, but while selling uses a scale which is 10 cm shorter than a normal metre scale. What is his gain? **☑ 2016 I**
(a) 20% (b) 21% (c) $22\frac{2}{9}\%$ (d) $23\frac{1}{3}\%$

> ANSWERS

1	a	2	b	3	c	4	a	5	b	6	a	7	a	8	a	9	c	10	d
11	d	12	c	13	c	14	c	15	c	16	c	17	b	18	c	19	c	20	b
21	b	22	d	23	c	24	c	25	b	26	c	27	c	28	d	29	b	30	b
31	d	32	d	33	b	34	b	35	a	36	c	37	b	38	c	39	c	40	c
41	c	42	a	43	c	44	b	45	c	46	a								

HINTS AND SOLUTIONS

1. (a) Given, selling price of article
 $= ₹ 247.50$ and gain $= \frac{25}{2}\%$
 $\therefore$ Cost price
 $= ₹ \left\{ \frac{100}{\left(100 + \frac{25}{2}\right)} \times 247.50 \right\}$ [Rule 1]
 $= ₹ \frac{100 \times 2 \times 247.50}{225} = ₹ 220$
2. (b) Total gain = SP - CP
 $= (840 - 720) = ₹ 120$
 $\therefore$ Gain percent $= \frac{120}{720} \times 100 = 16\frac{2}{3}\%$
3. (c) Given, SP = ₹ 110 and loss = 12%
 $\therefore$ Cost price $= ₹ \left(\frac{100}{88} \times 110 \right) = ₹ 125$
 Now, CP = ₹ 125, gain required = 8%
 $\therefore$ SP $= ₹ \left(\frac{(100 + 8)}{100} \times 125 \right) = ₹ 135$
4. (a) Let cost of one table and one chair be ₹ x and ₹ y , respectively.
 $\therefore$ Cost of 4 tables and 5 chairs
 $= ₹ 4x + 5y$
 $\therefore 4x + 5y = 1000$... (i)
 $\therefore$ Now, profit on 4 tables
 $= 4x \times 10\% = ₹ \frac{4x}{10}$
 Profit on 5 chairs $= 5y \times 20\% = ₹ y$
 $\therefore \frac{4x}{10} + y = 120$... (ii)
 Now, on solving Eqs. (i) and (ii), we get
 $x = ₹ 200$
5. (b) Refer to example 9.
6. (a) Refer to example 4.
7. (a) Let the selling price be ₹ 100.
 $\therefore$ Cost price $= \frac{SP \times 100}{100 + 10}$ [Rule 1]
 $= \frac{100 \times 100}{110} = ₹ \frac{1000}{11}$
 Now, if SP is ₹ 200.
 $\therefore$ Gain $= ₹ \left(200 - \frac{1000}{11} \right) = ₹ \frac{1200}{11}$
 Gain percent $= \frac{1200 / 11}{1000 / 11} \times 100 = 120\%$
8. (a) Given, selling price of an article
 $= ₹ 240$
 Loss = 4%
 $\therefore$ Cost price of an article for the loss of 4%
 $= ₹ \frac{240 \times 100}{96}$ [Rule 2]

Selling price of an article for a profit of 10%
 $= \frac{240 \times 100}{96} \times \frac{110}{100} = \frac{240 \times 110}{96}$
 $= ₹ 275$

9. (c) Let selling price of 1 dozen pencil be ₹ x .
 $\therefore$ Selling price of 8 dozen pencils = ₹ $8x$ and profit = ₹ x
 $\therefore$ Cost price of 8 dozen pencils
 $= 8x - x = ₹ 7x$
 $\therefore$ Gain, percent $= \frac{x}{7x} \times 100\% = 14\frac{2}{7}\%$
10. (d) Given, CP = ₹ 400, let selling price be ₹ x .
 Then, $400 + 20\%$ of $x = x$
 $\Rightarrow 400 + \frac{x}{5} = x \Rightarrow \frac{4x}{5} = 400$
 $\Rightarrow x = \frac{400 \times 5}{4} \Rightarrow x = ₹ 500$
 $\therefore$ Selling price = ₹ 500
11. (d) The cost price of table for person B
 $= 2000 + 6 \times \frac{2000}{100}$
 $= 2000 + 120 = ₹ 2120$
 Selling price for person B
 $= 2120 - \frac{2120 \times 5}{100}$
 $= 2120 - 106 = ₹ 2014$
12. (c) Let marked price be ₹ x .
 Selling price after 5% discount
 $= x - \frac{5}{100}x = \frac{19}{20}x$
 Profit = SP - CP $= \frac{19}{20}x - 380$
 Profit % $= \frac{\frac{19}{20}x - 380}{380} \times 100$
 $= \frac{\frac{19}{20}x - 380}{380} \times 100$
 $25 = \frac{\frac{19}{20}x - 380}{380} \times 100$
 $x = 475 \times \frac{20}{19} = ₹ 500$
13. (c) Refer to example 12.
14. (c) Refer to example 13.
15. (c) Case I Discount = 40 %
 $\Rightarrow$ Selling price = 60% of 1000
 $= \frac{60}{100} \times 1000 = ₹ 600$
 Case II Two successive discounts are of 35% and 5%.

- $\Rightarrow$ Selling price
 $= 1000 \times \left(1 - \frac{35}{100}\right) \left(1 - \frac{5}{100}\right)$
 [Rule 11]
 $= \frac{65}{100} \times \frac{95}{100} \times 1000 = ₹ 617.50$
 $\therefore$ Difference $= (617.50 - 600) = ₹ 17.50$
16. (c) List price of article be ₹ 100.
 Cost price for dealer
 $= \frac{(100 - 20)}{100} \times \frac{(100 - 10)}{100} \times 100$
 $= \frac{80 \times 90 \times 100}{100 \times 100} = ₹ 72$
 Money spent on transport
 $= \frac{10}{100} \times 72 = ₹ 7.20$
 $\therefore$ Total cost price $= 72 + 7.20 = ₹ 79.20$
 $\therefore$ Selling price
 $= \frac{(100 + 15)}{100} \times 79.20$
 $= \frac{115 \times 79.20}{100} = ₹ 91.08$
 17. (b) Given, cost of article = ₹ 200
 $\therefore$ Selling price of article = 95% of (90% of 200)
 $= \frac{95}{100} \times \frac{90}{100} \times 200 = ₹ 171$
 18. (c) Given, CP of 3 oranges of Ist variety
 $= ₹ 1$
 CP of 1 orange of Ist variety $= ₹ \frac{1}{3}$
 CP of 2 oranges of IInd variety = ₹ 1
 CP of 1 orange of IInd variety $= ₹ \frac{1}{2}$
 Total CP of 2 oranges of different variety
 $= \frac{1}{3} + \frac{1}{2} = ₹ \frac{5}{6}$
 Profit on 2 oranges of different variety
 $= 20\%$ of $\frac{5}{6} = \frac{20}{100} \times \frac{5}{6} = ₹ \frac{1}{6}$
 $\therefore$ SP of 2 oranges of different variety
 $= \frac{5}{6} + \frac{1}{6} = ₹ 1$
 Hence, SP of 12 oranges is ₹ 6.
 19. (c) Given, cost price of 1L = ₹ 8.50
 $\therefore$ Total CP of milk $= 28 \times 8.50 = ₹ 238$
 $\therefore$ Profit = 12.5% of 238
 $= \frac{12.5}{100} \times 238 = ₹ 29.75$

Let he added x L of water.

$$\therefore \text{Profit} = x \times 8.5 \Rightarrow 29.75 = x \times 8.5$$

$$\therefore x = 3.5 \text{ L}$$

20. (b) Refer to example 6.

$$\begin{aligned} \text{21. (b) Cost price of computer for Brajesh} \\ = 40000 - \frac{4}{100}(40000) = ₹ 38400 \end{aligned}$$

$$\begin{aligned} \text{Selling price of computer for Brajesh} \\ = ₹ 40320 \end{aligned}$$

$$\therefore \text{Profit amount} = (40320 - 38400) = ₹ 1920$$

$$\begin{aligned} \therefore \text{Profit percentage earned by Brajesh} \\ = \frac{1920}{38400} \times 100 = 5\% \end{aligned}$$

22. (d) Given, list price of a video cassette = ₹ 100

Let the rate of discount be $r\%$.

$$\begin{aligned} \text{Selling price of 3 video cassette} \\ = ₹ 274.50 \end{aligned}$$

$$\therefore \text{Selling price of 1 video cassette} = ₹ \frac{274.50}{3} = ₹ 91.50$$

$$\therefore 100 - \frac{r}{100} \times 100 = ₹ 91.50$$

$$\Rightarrow 100 - 91.50 = r \Rightarrow 8.50 = r$$

$$\therefore \text{Rate of discount} = 8.5\%$$

23. (c) CP of 1 lemon = $\frac{1}{2}$

$$\text{SP of 1 lemon} = \frac{3}{5}$$

$$\therefore \text{Gain \%}$$

$$= \frac{\frac{3}{5} - \frac{1}{2}}{\frac{1}{2}} \times 100 = \frac{6-5}{10} \times 2 \times 100 = 20\%$$

24. (c) Let the cost price be ₹ x .

$$\text{Marked price} = \frac{x \times 110}{100} = ₹ \frac{11x}{10}$$

$$\therefore \text{SP} = \frac{11x}{10} \times \frac{90}{100} = \frac{99x}{100}$$

$$\therefore \text{Required gain/loss per cent}$$

$$= \frac{\frac{99x}{100} - x}{x} \times 100 = -1\% \quad [\text{Rule 8}]$$

[negative sign i.e. loss]

25. (b) Let marked price of two sarees be ₹ x and ₹ y , respectively.

$$\therefore x - \frac{6x}{100} = 564 \Rightarrow \frac{94x}{100} = 564$$

$$\Rightarrow x = ₹ 600 \text{ and } y - \frac{y}{100} = 396$$

$$\Rightarrow \frac{99y}{100} = 396 \Rightarrow y = ₹ 400$$

$$\therefore \text{Total MP amount} = 600 + 400 = ₹ 1000$$

$$\begin{aligned} \text{Total amount after discount} \\ = 564 + 396 = ₹ 960 \end{aligned}$$

$$\therefore \text{Discount per cent}$$

$$= \frac{1000 - 960}{1000} \times 100 = \frac{40}{10}\% = 4\%$$

26. (c) Here, true weight = 1000

and gain = 25%

$$\Rightarrow 25 = \frac{1000 - \text{false weight}}{\text{false weight}} \times 100$$

$$\Rightarrow \frac{\text{false weight}}{4} = 1000 - \text{false weight}$$

$$\Rightarrow \text{false weight} = 1000 \times \frac{4}{5} = 800$$

Hence, he uses the weight 800 g.

27. (c) Let cost price of one scooter be ₹ x .

$$\begin{aligned} \therefore \text{Cost price of other scooter} \\ = ₹ (9000 - x) \end{aligned}$$

$$\begin{aligned} \therefore \text{Selling price of 1st scooter} \\ = x + \frac{25x}{100} = \frac{125x}{100} \end{aligned}$$

Also, selling price of 2nd scooter

$$\begin{aligned} &= (9000 - x) \left(1 - \frac{20}{100}\right) \\ &= (9000 - x) \left(\frac{80}{100}\right) \end{aligned}$$

Total selling price of both scooters

$$= \frac{125x}{100} + (9000 - x) \frac{80}{100}$$

$$\therefore \frac{125x}{100} + (9000 - x) \frac{80}{100} = 9000 \quad [\text{given}]$$

$$\Rightarrow \frac{45x}{100} + 7200 = 9000$$

$$\therefore x = 4000$$

$$\therefore \text{Cost price of 1st scooter} = ₹ 4000$$

$$\begin{aligned} \text{Hence, cost price of 2nd scooter} \\ = (9000 - x) = ₹ 5000 \end{aligned}$$

28. (d) Let the number of items be x .

Then, selling price of items = $60x$

Cost of material of items = $40x$

Overhead expenses = ₹ 3000

$$\therefore 60x - (40x + 3000) = 1000$$

$$\Rightarrow 20x = 4000$$

$$\therefore x = \frac{4000}{20} = 200$$

29. (b) I. Let CP = x

$$\text{Then, SP} = \frac{213x}{200}$$

$$\text{If SP} = \frac{213x}{200} + 1250, \text{ then gain} = 19\%$$

$$\Rightarrow 19 = \frac{\frac{213x}{200} + 1250 - x}{x} \times 100$$

$$\Rightarrow \frac{19x}{100} - \frac{213x}{200} = 1250$$

$$\Rightarrow x = 1250 \times 8 = 10000$$

$$\text{II. If SP} = \frac{213x}{200} + 2400 \text{ and gain}$$

= 19%, then

$$19 = \frac{\frac{213x}{200} + 2400 - x}{x} \times 100$$

$$\Rightarrow \frac{19x}{100} - \frac{13x}{200} = 2400$$

$$\Rightarrow x = 2400 \times 8 = 19200$$

Hence, I is correct but II is incorrect.

30. (b) Let CP = x

$$\text{then SP} = \frac{110x}{100} = \frac{11x}{10}$$

$$\text{Now, CP} = 96\% \text{ of } x = \frac{96x}{100} = ₹ \frac{24x}{25}$$

According to the questions

$$\text{SP} = ₹ \left(\frac{11x}{10} + 6 \right)$$

$$\therefore \left(\frac{11x}{10} + 6 \right) = 118 \frac{3}{4}\% \text{ of } \frac{24x}{25}$$

$$\Rightarrow \frac{11x + 60}{10} = \frac{475}{400} \times \frac{24x}{25} = \frac{57x}{50}$$

$$\Rightarrow 570x = 550x + 3000$$

$$\Rightarrow x = \frac{3000}{20} = 150$$

$$\therefore \text{CP} = ₹ 150$$

$$\therefore \text{SP} = \frac{11}{10} \times 150 = ₹ 165$$

Hence, I and II both are correct.

31. (d) Let CP of 8 quintal be ₹ x .

$$\therefore \text{CP of 1 quintal} = ₹ \frac{x}{8}$$

$$\therefore \text{SP of 3 quintal rice at 10\% profit}$$

$$= \frac{3x}{8} + \frac{3x}{8} \times \frac{1}{10} = \frac{3x}{8} + \frac{3x}{80} = \frac{33x}{80}$$

SP of 3 quintal rice without profit or

$$\text{loss} = ₹ \frac{3x}{8}$$

SP of 2 quintal rice at 5% loss

$$= \frac{2x}{8} - \frac{2x}{8} \times \frac{5}{100} = \frac{x}{4} - \frac{x}{4 \times 20}$$

$$= \frac{19x}{4 \times 20} = \frac{19x}{80}$$

$$\therefore \text{Total SP} = \frac{33x}{80} + \frac{3x}{8} + \frac{19x}{80}$$

$$= \frac{33x + 30x + 19x}{80} = \frac{82x}{80}$$

$$\therefore \text{Profit percentage} = \frac{\text{SP} - \text{CP}}{\text{CP}} \times 100\%$$

$$= \frac{\frac{82x}{80} - x}{x} \times 100\% = \frac{(82 - 80)x}{80x} \times 100\%$$

$$= \frac{2}{80} \times 100 = 2.5\%$$

32. (d) Let the CP of two articles be $3x$ and $5x$, respectively.

$$\therefore \text{SP of first article} = \frac{3x \times 70}{100} = \frac{21x}{10}$$

[Rule 2]

$$\text{and SP of second article} \\ = \frac{5x \times 120}{100} = 6x$$

$$\therefore \text{Total SP} \\ = 6x + \frac{21x}{10} = \frac{60x + 21x}{10} = \frac{81x}{10}$$

$$\therefore \text{Total CP} = 3x + 5x = 8x$$

$$\therefore \text{Profit} = \frac{81x}{10} - 8x = \frac{81x - 80x}{10} = \frac{x}{10}$$

$$\therefore \text{Overall percentage of gain percentage} \\ = \frac{\frac{x}{10} \times 100}{8x} = \frac{x \times 100}{10 \times 8x} = 1.25\%$$

33. (b) We know that,

$$\text{Net percentage discount} \\ = \frac{\text{Discount}}{\text{Cost price}} \times 100\% = \frac{1}{4} \times 100\% = 25\%$$

34. (b) We know that,

$$\text{Cost price} = \left(\frac{100}{100 - \text{Loss}\%} \right) \times \text{SP}$$

[Rule 2]

$$= \left(\frac{100}{100 - 15} \right) \times 136$$

$$= \frac{136 \times 100}{85} = ₹ 160$$

$\therefore$ Given, profit percentage = 15%

$$\therefore \text{Selling price (x)} \\ = \frac{160 \times (100 + 15)}{100} = \frac{160 \times 115}{100} = ₹ 184$$

So, option (b) is correct because $180 < x < 190$.

35. (a) Given, list price of television = ₹ 5000

and discount = 10%

$$\therefore \text{After discount television cost} \\ = 5000 - 5000 \times 10\% \\ = 5000 - \frac{5000 \times 10}{100} = ₹ 4500$$

Additional discount = 2%

$$\therefore \text{Actually price of television} \\ = 4500 - \frac{4500 \times 2}{100} = ₹ 4410$$

36. (c) $\therefore$ CP of 80 apples = ₹ 240

$$\therefore \text{CP of 1 apple} = \frac{240}{80} = ₹ 3$$

$\therefore$ CP of 20 apples = ₹ 3 $\times$ 20 = ₹ 60
To earn a profit of 20%, SP = 120% of 240 = ₹ 288

But he sold $\frac{1}{4}$ of his apples i.e. 20 apples for ₹ 60.

$$\therefore \text{SP of remaining 60 apples} \\ = (288 - 60) = ₹ 228$$

$$\therefore \text{SP of 1 apple} = \frac{228}{60} = ₹ 3.80$$

37. (b) Let the cost price of the article be ₹ x .

Given, profit percentage = 20%

$$\text{Now, } x + \frac{20x}{100} = 3600 \Rightarrow \frac{120x}{100} = 3600$$

$$\therefore x = ₹ 3000$$

Now, profit percentage when the article is sold for ₹ 3150

$$\frac{3150 - 3000}{3000} \times 100 = \frac{150}{3000} \times 100 = 5\%$$

38. (c) Let each lot of onion contains x kg. Then, total cost price of these two lots together = $10x + 15x = 25x$

Selling price of whole lot

$$= 15 \times (x + x) = 15 \times 2x = 30x$$

$$\therefore \text{Profit percentage} = \frac{30x - 25x}{25x} \times 100 \\ = \frac{5x}{25x} \times 100 = 20\%$$

Hence, the profit is 20%.

39. (c) Let the original price be ₹ x .

Since, at discount of 20% article costs ₹ 596.

$$\text{Then, } 596 = \frac{80}{100} \times x \Rightarrow x = \frac{596 \times 100}{80} \\ = ₹ 745$$

Hence, the original price of an article is ₹ 745.

40. (c) Given, cost price of 200 oranges = ₹ 1000

$$\therefore \text{Cost price of 1 orange} = \frac{1000}{200} = ₹ 5$$

$$\text{Selling price of 1 orange} = 5 \left(\frac{100 + 25}{100} \right) \\ = 5 \times \frac{125}{100} = ₹ 6.25$$

Now, in ₹ 6.25 number of oranges can be sold = 1

$$\text{In ₹ 100, number of oranges can be sold} \\ = \frac{1}{6.25} \times 100 = 16$$

Hence, 16 oranges can be sold in ₹ 100 for profit 25%.

41. (c) Let the cost price of article be ₹ x .

Discount = 20% and selling price = 24

[given]

$$\therefore x = 24 + 20\% \text{ of } x$$

$$\Rightarrow x = 24 + \frac{20}{100}x \Rightarrow x = 24 + \frac{x}{5}$$

$$\Rightarrow x - \frac{x}{5} = 24 \Rightarrow \frac{4x}{5} = 24$$

$$\Rightarrow x = \frac{24 \times 5}{4} = 30$$

$\therefore$ Cost price of an article = ₹ 30

Now, selling price after 30% discount

$$= 30 - 30\% \text{ of } 30 = 30 - \frac{30}{100} \times 30$$

$$= 30 - 9 = ₹ 21$$

42. (a) $\therefore$ Actual profit percentage $= \frac{\text{Fair weight} - \text{Unfair weight}}{\text{Unfair weight}} \times 100\%$

$$= \frac{1000 - 800}{800} \times 100\% = 25\%$$

43. (c) Let the cost price of an article be ₹ x . and selling price of an article = ₹ 96

[given]

$$\frac{x - 96}{x} \times 100 = \frac{1}{4}x$$

$$\Rightarrow x^2 - 400x + 38400 = 0$$

$$\Rightarrow x^2 - 160x - 240x + 38400 = 0$$

$$\Rightarrow (x - 160)(x - 240) = 0$$

$$\therefore x = 160 \text{ or } 240$$

Hence, the cost price of the article is either

₹ 160 or ₹ 240.

44. (b) Let CP of 1 L of milk be ₹ x .

$$\therefore \text{SP of 1 L of milk} \\ = x \times 120\% = ₹ 1.2x$$

Now, as in ₹ 1.2x, the quantity of milk sold = 1 L

$$\therefore \text{In ₹ } x, \text{ quantity of milk sold} \\ = \frac{1}{1.2x} \times x = \frac{5}{6} \text{ L}$$

According to the question,

CP of milk and SP of mixture are same, therefore in mixture, quantity of milk must be $\frac{5}{6}$ L.

Hence, the required percentage

$$= \frac{5}{6} \times 100\% = \frac{250}{3}\%$$

45. (c) Refer to example 12.

46. (a) Let the amount of cloth = 100 cm

Then, amount of cloth purchased

$$= 110 \text{ cm}$$

and amount of cloth sold = 90 cm

$$\therefore \text{Gain \%} = \frac{110 - 90}{100} \times 100\%$$

$$= \frac{20}{100} \times 100\% = 20\%$$